

PRISCILLA MALOMBE NZULA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Scientist and Analyst with over 5 years of experience spanning corporate banking and independent data science practice. At Equity Bank, managed credit analytics for a KES 450 billion loan portfolio across 30,000 clients. Since September 2024, has been building end-to-end data science projects across manufacturing, HR, retail, and healthcare domains, delivering production-ready machine learning models, live dashboards, and automated pipelines. Skilled in Python, SQL, machine learning, and cloud deployment.

TECHNICAL SKILLS

Programming and Libraries: Python, NumPy, Pandas, Scikit-learn, TensorFlow, Keras, XGBoost, Joblib

Machine Learning: Predictive Modeling, Random Forest, Isolation Forest, Feature Engineering, SMOTE, Deep Learning, Transfer Learning, NLP

Database and SQL: MySQL, Supabase, Querying, ETL Processes, Database Design, Stored Procedures

Data Visualization: Streamlit, Plotly, Matplotlib, Seaborn, Chart.js, Power BI, Tableau, Excel

Deployment and Cloud: GitHub Actions, Streamlit Community Cloud, Netlify, FastAPI, CI/CD Pipelines

Tools and Platforms: Jupyter Notebook, VS Code, Google Colab, Git, GitHub

WORK EXPERIENCE

Data Scientist

September 2024 – Present

Freelance | Nairobi, Kenya

- Delivered a manufacturing analytics project for turbofan engine maintenance, enabling the business to identify critical engines and schedule interventions in advance rather than reacting to unexpected failures.
- Built an HR analytics solution that gave a people team early visibility into which employees were most likely to leave, surfacing the departments and workload patterns driving the highest turnover.
- Produced a retail analytics report for a fashion business that clarified which customer segments and product categories were driving the majority of revenue, informing inventory and marketing priorities.
- Developed a telecoms churn model that identified customers most likely to cancel their contracts, giving the retention team a prioritized list to act on before attrition occurred.
- Designed and implemented two production database systems covering e-commerce and library operations, replacing ad hoc data handling with structured schemas and automated data integrity checks.

Relationship Officer, Credit Monitoring and Collections

November 2021 – August 2024

Equity Bank Limited | Nairobi, Kenya

- Analyzed financial data and loan repayment patterns for a portfolio of 30,000 clients across Corporate, SME, and Retail segments valued at KES 450 billion, identifying trends and default risks.
- Built and maintained Excel dashboards to monitor portfolio performance, track overdue payments, and present key risk indicators to management.
- Applied data-driven insights from customer payment histories and financial statements to support account restructuring and creditworthiness assessments, contributing to a 30% reduction in monthly provisioning and account slippage.
- Extracted and synthesized client data from core banking systems (FinOne, Finacle) to ensure accurate reporting and compliance with credit monitoring protocols.

Relationship Officer, Cash

August 2019 – October 2021

Equity Bank Limited | Nairobi, Kenya

- Processed and reconciled high-volume client transactions, ensuring accuracy and compliance with financial regulations.
- Monitored client account activity to identify irregular patterns and flag potentially fraudulent transactions for investigation.
- Compiled daily transaction reports to track branch performance metrics, providing data that supported management decision-making.

PROJECTS

Turbofan Engine Predictive Maintenance | Python, Random Forest, MySQL, Supabase, Streamlit, GitHub Actions

[Live dashboard](#)

- Trained a Random Forest regressor on 20,631 NASA CMAPSS sensor readings to predict engine Remaining Useful Life, achieving $R^2 = 0.969$ and RMSE = 7.2 cycles. Engineered rolling average and standard deviation features over a 20-cycle window across four key sensors to capture degradation trends.
- Connected the model to a Supabase cloud database, built a live Streamlit dashboard showing fleet health across 100 engines, and automated daily data refresh and maintenance alerts via a GitHub Actions cron workflow.

HR Analytics: Employee Attrition Prediction | Python, Random Forest, XGBoost, FastAPI, SMOTE
[Link](#)

- Trained and compared Random Forest and XGBoost classifiers on 29,524 employee records, applying SMOTE to correct class imbalance. The best model achieved 0.87 ROC-AUC and caught 75% of at-risk employees before they resigned.
- Found a 33% company-wide attrition rate concentrated in Finance, Sales, and Operations, with number of projects, monthly hours, and satisfaction score as the top three predictors. Deployed the model as a FastAPI REST endpoint for real-time scoring.

StyleSense: Retail Sales and Customer Analytics | Python, Pandas, RFM Segmentation, Plotly
[Link](#)

- Segmented 12 months of retail transaction data using RFM analysis, classifying customers into Champions, Loyal, At-Risk, and Lost groups. Found that the top 20% of customers generated over 80% of total revenue.
- Forecasted seasonal demand patterns and identified the highest-margin product categories, producing a ranked list of restocking priorities and markdown recommendations backed by transaction-level data.

EDUCATION

BSc Mathematics and Computer Science	September 2015 – October 2019
<i>Multimedia University of Kenya</i>	

CERTIFICATIONS

Data Science Certification, Moringa School	August 2023 – June 2024
Data Protection Certification, CIPIT Strathmore University	October 2024